

Alloys

1. Galvanized iron pipes have a coating of :

- (a) Zinc (b) Mercury
(c) Lead (d) Chromium
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

Galvanization is the process of applying a protective zinc coating on steel or iron to prevent them from rusting and oxidation. The zinc forms a barrier between atmospheric oxygen and the underlying iron or steel. It does this by transferring electrons and oxidizing more quickly than iron. This rapid zinc oxidation prevents ferrous metals from rusting until the zinc has exhausted its free-electron capacity or the protective coating has worn away.

2. Alloy of which metal is used to make aeroplane and parts of the compartment of the train?

- (a) Copper (b) Iron
(c) Aluminium (d) None of these

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Low density and strength make aluminium ideal for construction of aircraft, lightweight vehicles and ladders. An alloy of aluminium called Duralumin is often used instead of pure aluminium because of its improved properties. Easy shaping and corrosion resistance make aluminium a good material for drinking cans and roofing materials.

3. Mercury-metal mixture is –

- (a) High colour alloy
(b) Carbon mixed alloy
(c) Mercury mixed alloy
(d) Highly resisting alloy for malfriction

39th B.P.S.C. (Pre) 1994

Ans. (c)

An amalgam is a substance formed by the reaction of Mercury with another metal. Almost all metals can form amalgams with Mercury, the notable exception being Iron.

4. Stainless steel is an alloy of

- (a) iron and copper (b) iron and zinc
(c) iron and chromium (d) iron and graphite

38th B.P.S.C. (Pre) 1992-93

Ans. (c)

Stainless steel is a corrosion - resistant alloy of iron, chromium and, in some cases, nickel and other metal. Hence option (c) is the correct answer.

5. Brass is an alloy of –

- (a) Copper and Nickel (b) Nickel and Zinc
(c) Copper and Zinc (d) Iron and Nickel

47th B.P.S.C. (Pre) 2005

Ans. (c)

Brass consists 68-71% Copper and rest is Zinc. Bronze consists 88% Copper and 12% Tin. German silver has almost 50% Copper. Gunmetal consists 85% Copper, 5% Tin, 5% Lead, 5% Zinc.

6. Which among the following are constituents of brass?

- (a) Zinc and copper (b) Iron and zinc
(c) Copper and nickel (d) Iron and copper
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

See the explanation of the above question.

7. Brass is formed by mixing one volume part of zinc (density 7100 kg/m) with two volume parts of copper (density 8900 kg/m³). The density of brass is -

- (a) 8000 kg/m³ (b) 8300 kg/m³
(c) 9500 kg/m³ (d) 9800 kg/m³

46th B.P.S.C. (Pre) 2004

Ans. (b)

3 brass = 1 part zinc + 2 parts copper
= 7100 + 2 × 8900 = 24900
1 part brass = 24900/3 = 8300 kg/m³

8. To make the steel hard requires increase in –

- (a) The quantity of Carbon
(b) The quantity of Manganese
(c) The quantity of Silicon
(d) The quantity of Chromium

42nd B.P.S.C. (Pre) 1997

Ans. (a)

Steels are alloys of Iron and other elements primarily Carbon and are widely used in construction and other area. Carbon mixed in Iron as a hardening agent which strengthens Iron by distorting its crystal lattice. This distortion results in hardening.

9. Stainless steel is an alloy of :

- (a) Iron and nickel (b) Iron and chromium
(c) Copper and chromium (d) Iron and zinc
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

Stainless steel (stain -free iron) is an alloy whose constituent elements generally are as follows :

Iron	-	89.4%
Chromium	-	10.0%